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Biochemical and histopathological evaluation of atrazine toxicity and herbal protection by *Ocimum sanctum*

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Abstract- Atrazine is a widely used triazine herbicide known for its persistence in environmental systems and its potential to induce toxic effects in living organisms. The present study evaluates the impact of sub-chronic atrazine exposure on biochemical and oxidative stress parameters in *Mus musculus* and examines the protective role of *Ocimum sanctum*. Experimental animals were divided into four groups: control, atrazine-treated, and two groups treated with atrazine along with Tulsi extract (100 mg/kg and 200 mg/kg). Atrazine exposure resulted in significant metabolic disturbances, including hyperglycemia, hypercholesterolemia, reduced plasma protein levels, increased acid phosphatase activity, and elevated oxidative stress markers. Co-treatment with Tulsi extract improved these alterations in a dose-dependent manner, with the higher dose showing near normalization. These findings indicate that Tulsi possesses strong antioxidant and cytoprotective properties and may serve as a natural therapeutic agent against pesticide-induced toxicity.

Keywords: Atrazine, *Ocimum sanctum*, oxidative stress, biochemical parameters, hepatotoxicity, nephrotoxicity

INTRODUCTION

The increasing dependence on chemical herbicides in agriculture has significantly improved crop productivity, yet it has also introduced serious environmental and health concerns. Atrazine, a member of the triazine class of herbicides, is widely used for controlling broadleaf weeds in crops such as maize, sorghum, and sugarcane. Its extensive use is linked to its effectiveness and low cost.[1]

Atrazine is chemically stable and persists in soil and water for extended periods, making it a common contaminant in groundwater and surface water systems.[2] Continuous exposure, even at low concentrations, poses risks to both ecosystems and human health. Studies have shown that atrazine interferes with endocrine function, particularly affecting the hypothalamic-pituitary-gonadal axis, leading to reproductive and developmental disturbances.[2,3]

At the cellular level, atrazine promotes the generation of reactive oxygen species, resulting in oxidative stress and damage to biomolecules such as lipids, proteins, and DNA.[3] These effects are often reflected in altered biochemical parameters including glucose, cholesterol, and protein levels.

Medicinal plants have been explored for their protective roles against toxicants. *Ocimum sanctum* (Tulsi) is a well-known medicinal herb with antioxidant, anti-inflammatory, and hepatoprotective properties. Its phytoconstituents, such as eugenol and flavonoids, play a

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key role in neutralizing free radicals and restoring cellular balance.[4,5]

The present study investigates the biochemical and oxidative stress alterations induced by atrazine exposure and evaluates the protective efficacy of Tulsi in a controlled experimental model.

MATERIALS & METHODS

Experimental Animals and Design

Healthy albino mice (*Mus musculus*) were used for the present study and maintained under standard laboratory conditions with controlled temperature, light cycle, and free access to food and water. The experiment was conducted using a 28-day sub-chronic exposure model to assess the toxicological effects of atrazine. The animals were randomly divided into four groups: Group A served as the control, Group B received atrazine treatment, Group C received atrazine along with *Ocimum sanctum* (Tulsi) extract at a dose of 100 mg/kg body weight, and Group D received atrazine along with Tulsi extract at 200 mg/kg body weight. All treatments were administered orally to simulate natural exposure conditions.

Preparation of Plant Extract

Fresh leaves of *Ocimum sanctum* were collected and washed thoroughly to remove dust and impurities. The cleaned leaves were boiled in distilled water for a short duration to obtain an aqueous extract, which was then filtered using standard filter paper to remove solid residues. The resulting extract was used for oral administration, as this preparation method closely reflects traditional medicinal practices.

Biochemical Analysis

At the end of the experimental period, blood samples were collected for biochemical analysis. Key parameters were measured using established standard methods. Blood glucose levels were determined using the Anthrone method, while serum cholesterol was estimated using Zak's method. Total protein content was measured by the Lowry method, and acid phosphatase (ACP) activity was assessed as an indicator of cellular damage and lysosomal integrity.

Oxidative Stress Assessment

To evaluate oxidative stress, tissue samples were analyzed for several important biomarkers. Lipid peroxidation was assessed by measuring malondialdehyde (MDA) levels. Antioxidant defense status was evaluated by determining the activities of superoxide dismutase (SOD) and catalase (CAT), along with levels of glutathione (GSH) and glutathione peroxidase (GPx). These parameters provided insight into the balance between oxidative damage and the antioxidant defense system in the experimental animals.

Table 1. Experimental Design and Treatment Groups

Group	Treatment Description	Dose
Group A	Control	—
Group B	Atrazine only	Standard dose
Group C	Atrazine + Tulsi	100 mg/kg
Group D	Atrazine + Tulsi	200 mg/kg

Table 2. Effect of Atrazine and Tulsi on Biochemical Parameters

Parameter	Control	Atrazine	Atrazine + Tulsi (100 mg/kg)	Atrazine + Tulsi (200 mg/kg)
Blood Glucose (mg/dL)	92.4 ± 3.1	136.8 ± 4.5	112.7 ± 3.8	98.6 ± 2.9
Cholesterol (mg/dL)	126.2 ± 4.2	172.5 ± 5.6	143.7 ± 4.8	132.4 ± 4.1
Plasma Protein (g/dL)	7.42	5.86	6.72	7.11
ACP (IU/mg protein)	12.4	20.8	15.6	13.2

Table 3. Effect on Body Weight Gain (%)

Group	Body Weight Gain (%)
Control	20.3
Atrazine	3
Atrazine + Tulsi (100 mg/kg)	12.1
Atrazine + Tulsi (200 mg/kg)	16.8

RESULTS

General Observations and Body Weight

Atrazine-treated animals showed signs of toxicity including lethargy, reduced food intake, and decreased locomotor activity. Body weight gain was significantly reduced compared to controls. Tulsi treatment improved physical activity and body weight in both treatment groups, with better recovery observed at 200 mg/kg.

Blood Glucose

Atrazine exposure caused a marked increase in blood glucose levels, indicating impaired glucose metabolism. Tulsi treatment reduced glucose levels toward normal values, suggesting improved metabolic regulation.

Serum Cholesterol

Serum cholesterol levels were significantly elevated in atrazine-treated animals, reflecting disturbance in lipid metabolism. Tulsi administration reduced cholesterol levels in a dose-dependent manner.

Plasma Protein and ACP Activity

Atrazine exposure decreased plasma protein levels and increased ACP activity, indicating liver dysfunction and cellular damage. Tulsi treatment restored these parameters toward normal values.

Oxidative Stress and Histopathology

Atrazine increased lipid peroxidation and reduced antioxidant enzyme levels. Histopathological examination revealed liver necrosis and kidney damage. Tulsi treatment improved antioxidant status and preserved tissue architecture.

DISCUSSION

Atrazine exposure produced clear signs of systemic toxicity, affecting metabolic, biochemical, and cellular processes. The observed reduction in body weight reflects impaired nutrient utilization and metabolic stress, which has been reported in earlier toxicological studies.[6]

Elevated blood glucose levels suggest disruption of glucose homeostasis, possibly due to oxidative damage to pancreatic β-cells or altered insulin signaling pathways. Similar findings have been documented in pesticide-induced metabolic dysfunction.[7] The increase in serum cholesterol indicates altered lipid metabolism, which may arise from hepatic dysfunction and oxidative damage to liver tissues.[6]

The decline in plasma protein levels is indicative of impaired protein synthesis in the liver. This may result from damage to hepatocytes, which play a central role in protein metabolism. Elevated ACP activity further supports the presence of cellular and lysosomal damage.

Oxidative stress appears to be a key mechanism underlying atrazine toxicity. Increased MDA levels reflect enhanced lipid peroxidation, while reduced levels of SOD, CAT, GPx, and GSH indicate depletion of the antioxidant defense system.[3] This imbalance leads to structural and functional damage in tissues, as confirmed by histopathological observations.

Tulsi supplementation showed a strong protective effect against these alterations. Its phytochemical constituents, including eugenol and flavonoids, are known to scavenge free radicals and enhance antioxidant enzyme activity.[4,5] The restoration of biochemical parameters suggests improved liver function and metabolic stability. Tissue recovery observed in histological analysis further confirms its protective role.

The higher dose of Tulsi (200 mg/kg) demonstrated greater efficacy, suggesting a dose-dependent response. This aligns with previous studies showing enhanced therapeutic effects of medicinal plants at higher concentrations within safe limits.[8,9]

These findings support the use of *Ocimum sanctum* as a natural protective agent against pesticide-induced toxicity and highlight its potential role in reducing oxidative stress and improving organ function.

CONCLUSION

The present study clearly demonstrates that sub-chronic exposure to atrazine induces significant biochemical and physiological disturbances in *Mus musculus*. These alterations were evident through elevated blood glucose and cholesterol levels, reduced plasma protein concentration, increased acid phosphatase activity, and pronounced oxidative stress marked by enhanced lipid peroxidation and depletion of antioxidant enzymes. Histopathological observations further confirmed liver and kidney damage, indicating systemic toxicity.

Exposure to *Ocimum sanctum* extract showed a strong protective effect against atrazine-induced toxicity. Tulsi treatment led to restoration of biochemical parameters toward normal levels, improvement in antioxidant enzyme activity, and reduction in oxidative damage. The higher dose (200 mg/kg) was more effective, showing near-complete recovery in several parameters and indicating a dose-dependent response.

These findings suggest that the toxic effects of atrazine are primarily driven by oxidative stress and metabolic disruption, while *Ocimum sanctum* exerts protective effects through its antioxidant and cytoprotective properties. The study supports the potential use of Tulsi as a natural therapeutic option for reducing pesticide-induced toxicity and aiding physiological recovery.

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